

## Experiment # 11

### Understanding of Transfer Learning

#### Objective/s:

- What is Transfer learning
- How to use a pre trained model on our own dataset?
- Lab Tasks

#### Theory:

Transfer learning is a machine learning technique where a model developed for one task is reused as the starting point for a model on a second task. This approach leverages the knowledge gained from a previously learned task and applies it to a new but related task. Transfer learning is particularly useful when the new task has limited labeled data, as it allows the model to benefit from the rich features learned from the larger dataset of the initial task.

#### Here are some key aspects of transfer learning:

1. **Pre-trained Models:** Transfer learning often involves using pre-trained models that have been trained on large datasets (like ImageNet for image classification tasks). These pre-trained models have already learned useful features that can be transferred to the new task.
2. **Fine-tuning:** After transferring the pre-trained model to the new task, fine-tuning involves training the model on the new task-specific data. This process adjusts the model's parameters slightly to better fit the new task.
3. **Feature Extraction:** In some cases, transfer learning is used to extract features from the new data using the pre-trained model, and then a new model (e.g., a simple classifier) is trained on these extracted features.
4. **Application Domains:** Transfer learning is widely used in various domains such as computer vision (e.g., image classification, object detection), natural language processing (e.g., sentiment analysis, machine translation), and more.

## Example Workflow

1. **Select a Pre-Trained Model:** Choose a model that has been pre-trained on a large dataset similar to your target task.
2. **Replace the Output Layer:** Modify the output layer of the pre-trained model to match the number of classes in your target task.
3. **Train the Model on New Data:** Train the modified model on your new, task-specific dataset. This can involve freezing some layers of the pre-trained model to preserve learned features and only training the new layers, or fine-tuning the entire model.

## Benefits

- **Improved Performance:** Models can achieve higher accuracy with less data.
- **Faster Training:** Reduces the time and computational resources required for training.
- **Overcome Data Scarcity:** Useful for tasks with limited labeled data.

## Lab Task:

Consider you want to build a model to classify types of flowers but you only have a small dataset. You can use a pre-trained model like VGG16, trained on ImageNet, replace its final layer to output the number of flower classes, and then fine-tune this model on your flower dataset. This approach takes advantage of the robust features learned by VGG16 from the large ImageNet dataset, improving the performance of your flower classification task.